

BASIC OS CONCEPTS

Condensed Study Notes

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BASIC OPERATING SYSTEM CONCEPTS

Imagine your computer is like a busy office. The Operating System (OS) is the office manager. It's the main program that makes sure everything runs smoothly and all the different parts of the computer work together.

Think of the OS as a conductor of an orchestra. The computer has many different instruments (like the screen, keyboard, and storage) and musicians (software programs). The conductor (OS) tells everyone when to play, what to play, and makes sure they all play in harmony.

The OS has a few really important jobs:

- **Managing Resources:** This is like the office manager deciding who gets to use which meeting room or which phone. The OS decides which programs get to use the computer's 'thinking power' (processor), 'short-term memory' (RAM), and other devices. It makes sure programs don't get in each other's way.
- **Providing a User Interface:** This is like the receptionist at the office who helps people find their way around and interact with the office. The OS gives you a way to interact with the computer, whether it's through clicking icons on a screen (graphical user interface) or typing commands (command-line interface). This is how you tell the computer what you want it to do.
- **Running Applications:** These are like the individual workers in the office, each doing a specific job (like a word processor, a web browser, or a game). The OS helps these applications start up, run, and finish their tasks efficiently.

Definition

We've already talked about the Operating System (OS) being the main manager of your computer.

Think of the OS like a really good event planner for a big party. The party has guests (users), activities (applications), and the venue itself (computer hardware). The event planner's job is to make sure everyone and everything works together smoothly so the party is a success.

More specifically, the OS is a type of software (programs on your computer) that acts as a go-between. It helps you, the programs you want to run, and the actual physical parts of your computer (like the screen, keyboard, and processor) to talk to each other.

It's essentially the communication bridge that lets all the different pieces of the computer system coordinate and work together harmoniously. The main goal is to make sure the whole computer system runs well, efficiently, and is easy for you to use.

Definition (contd)

Imagine your computer is like a busy restaurant.

1. **The OS as a Foundation for Apps:** Just like a restaurant needs a solid kitchen and well-trained staff before it can serve any dishes, your computer needs an Operating System (OS) to be a base for all the applications you use (like games, web browsers, or word processors). Without the OS, these applications wouldn't have anything to run on.

2. **The OS as a Middleman:** The OS also acts as a helpful go-between. It stands between you (the user) and the actual computer parts (the hardware – like the screen, keyboard, and processing chips). This means you don't have to know how to directly tell the hardware what to do; you just interact with the OS, and it handles the communication.

3. **Different OSs for Different Jobs:** Think about different types of restaurants – a fancy French restaurant will operate very differently from a quick-service burger joint. Similarly, different operating systems are designed for different kinds of computers and tasks. The main purpose or 'goal' of a computer system (like being super fast for gaming, or very secure for banking) will heavily influence how its OS is built and how it gets things done.

Computer System Components and Interaction

Imagine a computer system like a bustling office. Different parts work together to get things done, and there's a central manager (which we learned is the Operating System!) making sure everything runs smoothly.

This section is about showing you a picture, or a diagram, of all these different parts in the office and how they talk to each other to make the computer work.

It's like looking at a map of the office to see where the filing cabinets (where information is stored), the desks (where people work), and the communication lines (how they send messages) are, and how they're all connected.

We'll see how all the different pieces of the computer fit together and interact to carry out tasks.

Functions of the OS

Imagine your computer is a busy office. The Operating System (OS) is like the office manager, making sure everything runs smoothly and efficiently.

Here are the main jobs the OS does:

- **Resource Management:** Think of computer resources like the office supplies (pens, paper), the meeting rooms, and the electricity. These are limited, and the OS makes sure they are used wisely so the office can perform at its best. Resources include things like the CPU (the main worker), memory (where things are temporarily stored), network connection, storage space, and devices like printers.
- **Process Management:** A 'process' is just a program that's currently running. In our office analogy, these are like different tasks or projects happening at the same time. The OS decides which task gets to use the main worker (the CPU) and for how long, ensuring each task gets enough attention to be

completed. It also keeps track of what state each task is in (e.g., waiting, working) and assigns the CPU to a task when needed, and takes it away when the task is finished.

- **Memory Management:** This is like managing the desks and filing cabinets in the office. The OS manages the computer's 'main memory' (also called RAM). This is a fast storage area where programs need to be loaded before they can be run. The OS decides which part of the memory is given to which running task, and when a task is done or needs to wait for something else, the OS takes that memory back. It keeps track of which parts of memory are being used and which are free.

- **I/O Management:** This is about handling all the devices that send information in (Input) or get information out (Output), like keyboards, mice, printers, and screens. The OS keeps a list of all these devices and assigns a specific program, called an 'Input/Output controller', to manage each one. It decides which task gets to use a particular device and for how long, making sure devices are used effectively and are freed up when no longer needed.

- **Storage Management:** This is like managing the main library or warehouse of the office, where long-term information is kept. The OS handles how data is stored on devices like hard drives. This includes deciding how to save data, how users can find and access it, creating and organizing files and folders, reading and writing information, and copying data from one place to another within the storage.

History of operating systems

Let's take a trip back in time to see how computers and their managing software, the operating systems, have evolved!

The Very Beginning (First Generation Computers)

Imagine a time when computers were brand new, and not many people knew how they worked. In fact, the very first digital computer was designed by Charles Babbage, but it didn't quite work as planned because the technology of the day wasn't ready for his amazing ideas.

- Back then, a single team of people would design, build, program, run, and fix each computer. It was like a small, dedicated group managing a whole workshop.
- These early machines used something called **vacuum tubes**, which are like old-fashioned light bulbs, and some of them could be programmed.
- Operating systems as we know them didn't exist yet. Programmers would actually sign up for specific times to use the computer directly. It was like booking a spot in a workshop to use a special tool.
- The very first digital computer, by Charles Babbage, didn't have an operating system at all.

Getting More Organized (Second Generation Computers - Transistors and Batch Systems)

Things started to change with new technology. Instead of bulky vacuum tubes, computers began using **transistors**. Think of transistors as much smaller, more reliable versions of those old light bulbs, making computers smaller and more efficient.

- These new machines were called **mainframes**, which were very large and expensive computers.
- Because these machines were so costly, people wanted to make sure they were used as much as possible. This led to the idea of **batch systems**.

Imagine you have a pile of chores to do. Instead of doing them one by one as they come up, you gather all similar chores together and do them all at once. That's kind of how batch systems worked!

- Programmers would write their instructions on paper and then punch them onto **punch cards**. Think of these like old-school digital tickets.
- These punch cards were given to an operator who would collect a bunch of them together, forming a "batch." This batch might be collected for about an hour.
- The cards were then read onto a **magnetic tape**, which is like a very old, long ribbon for storing information.
- A special program, which was an early version of today's operating system, would then read the first job from the tape, run it, and save the results onto another tape for later printing.
- This was a big improvement because it meant the expensive computer wasn't sitting idle between jobs.

More Power and Smarter Systems (Third Generation Computers - ICs and Multiprogramming)

Computers continued to evolve with the invention of **Integrated Circuits (ICs)**, which are like tiny chips containing many electronic components. This made computers even smaller, faster, and more powerful.

- Computer makers started building different types of computers: one for doing lots of calculations (scientific purposes) and another for business tasks that involved more input and output of data (commercial purposes).
- Scientific computers were great at doing math and spent most of their time calculating, using the CPU a lot. Commercial computers did less calculation and spent more time dealing with getting information in and out (I/O).
- Before this, when a computer was busy getting information in or out (I/O), the **CPU** (the Central Processing Unit, the computer's brain) would just wait around doing nothing. This was a waste of its power.

This is where **multiprogramming** comes in! Think of it like a chef who, while waiting for something to bake in the oven (I/O), starts chopping vegetables for the next dish (using the CPU).

- With multiprogramming, if one task was waiting for information (I/O), another task could use the CPU to do its work. This kept the CPU much busier and made the computer more efficient.
- Eventually, companies like IBM created computers that could handle both scientific and commercial tasks. This meant operating systems needed to be able to do multiprogramming effectively.
- Over time, **time-sharing operating systems** were also developed. This is like multiple people taking turns using a single computer, with each person feeling like they have their own dedicated machine.

Fourth and fifth generation computers

Imagine your computer's operating system (OS) as the conductor of an orchestra. In the fourth and fifth generations of computers, this conductor really started to take shape into the sophisticated manager we know today.

- **Single Tasking OS:** Think of a very simple orchestra where the conductor can only lead one instrument at a time. This is like early operating systems, such as MS-DOS, which could only handle

one job or application at a time. If you were writing a document, the OS was focused solely on that task.

- **Multitasking OS:** Now, imagine a more complex orchestra where the conductor can seamlessly switch between leading different sections – the strings, then the brass, then the percussion. This is like the multitasking operating systems we use on PCs today, like Mac and Windows. They can manage and work on several tasks or applications at the same time, making our computers feel much more responsive.

- **Mobile Computing:** The fifth generation brought about a revolution in how we use computers. Think about your smartphone or tablet. These portable devices came with their own specialized operating systems, designed to fit into a smaller, mobile package. This allowed us to have powerful computing capabilities right in our pockets!

Types of operating systems

We've seen that Operating Systems (OS) are like the conductors of a computer orchestra, making sure everything runs smoothly. Now, let's explore some of the different "sections" or "sections" of this orchestra, which are the different types of OS designed for specific jobs.

- **Batch Operating Systems:**

Imagine you have a big pile of laundry to do. Instead of doing one sock, then one shirt, then another sock, you might sort your laundry into piles: whites, colors, delicates. You then wash all the whites together, then all the colors, and so on. This is similar to how batch operating systems work.

- **No direct user interaction:** You don't tell the computer "do this now." Instead, you prepare "jobs" (sets of instructions and data) beforehand.

- **Grouping jobs:** The OS groups jobs that have similar needs together into "batches."

- **Processing in batches:** The computer then processes these batches one after another without interruption or needing specific commands from you for each job.

- **Good for repetitive tasks:** This is useful when you have a lot of similar, large tasks to do repeatedly, especially if many people are using the system.

- **Multiprogramming OS:**

Think of a busy chef in a kitchen who is waiting for a pot to boil. Instead of just standing there, the chef uses that waiting time to chop vegetables for another dish. Multiprogramming OS does something similar with the computer's "brain," called the CPU (Central Processing Unit).

- **Better resource management:** The main goal is to keep the CPU busy and not let it sit idle.

- **Multiple jobs in memory:** Several "jobs" (programs or tasks) are loaded into the computer's "working memory" (RAM) at the same time.

- **Switching between jobs:** When one job needs to wait for something, like getting data from a storage device (this waiting is called I/O - Input/Output), the OS quickly switches the CPU's attention to another job that's ready to run. This way, the CPU is always working on something.

- **Multitasking OS:**

This is like being able to listen to music, chat with a friend online, and write a document all at the very same time on your computer. Multitasking OS allows you to do this.

- **Simultaneous tasks:** You can perform more than one computer "task" (a running program or process) at the same time.
- **Time sharing:** Each task gets a tiny slice of the CPU's time, called a "time quantum." The OS rapidly switches between these tasks, giving each a turn.
- **Process queue:** Tasks are placed in a line (a "queue") based on when they start. They take turns running for their time quantum.
- **Context switching:** When the OS switches from one task to another, it needs to quickly save the state of the first task so it can resume later, and load the state of the next task. This rapid saving and loading is called "context switching."
- **Multithreading:** This is a way to break down a single task into smaller parts that can run somewhat independently, allowing for more efficient multitasking within that task itself.

- **Multiprocessing OS:**

Imagine you have a huge construction project and instead of just one foreman overseeing everything, you have several foremen working together, each managing a different part of the project. Multiprocessing OS is for computers that have more than one CPU.

- **Multiple processors working together:** This OS is designed to make computers with multiple CPUs run faster and more efficiently by sharing the workload among them.
- **Increased speed and reliability:** Having multiple CPUs working together significantly speeds up how quickly tasks can be completed, and if one CPU has a problem, the system can often keep running because others are still available.

- **Real-time OS (RTOS):**

Think about a surgeon performing a delicate operation. Every action needs to happen at the exact right moment, with no delays. This is the essence of a real-time OS.

- **Time-bound applications:** These OS are designed for applications where timing is extremely critical.
- **Soft real-time:** In a "soft" real-time system, tasks are expected to finish within a certain timeframe, but a slight delay might be acceptable.

Hard real-time: In a "hard" real-time system, tasks must* be completed at their exact scheduled time. Missing a deadline can cause the entire system to fail.

- **Used in critical systems:** They are often used in applications like medical equipment, industrial control systems, and scientific experiments where precision timing is crucial.

Conclusion

We've covered quite a bit about Operating Systems (OS) so far!

Think of the OS like a really organized librarian for your computer. This librarian helps you find the books (programs) you want to read, makes sure you can use the library's resources (like the computer's memory and processing power) efficiently, and keeps track of all the different sections and how they work together.

Here's a quick recap of what we've explored:

- **What an OS does (its functions):** We looked at how the OS manages all the different parts of your computer, making sure everything runs smoothly and you can get your tasks done.
- **Different kinds of OS (types):** Just like libraries have different sections for different kinds of books, there are various types of OS designed for specific purposes, whether it's for a single user, many users, or even for machines that need to react instantly.
- **How OS came to be (history):** We saw how operating systems have evolved over time, from very simple beginnings to the complex systems we use today, getting smarter and more capable with each generation.
- **What an OS is (definition):** We established that the OS is the essential software that acts as a bridge, connecting you and your applications to the computer's hardware.